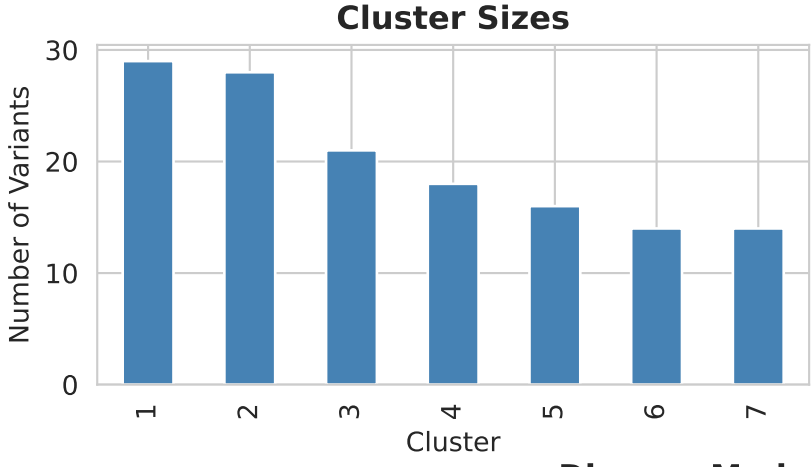
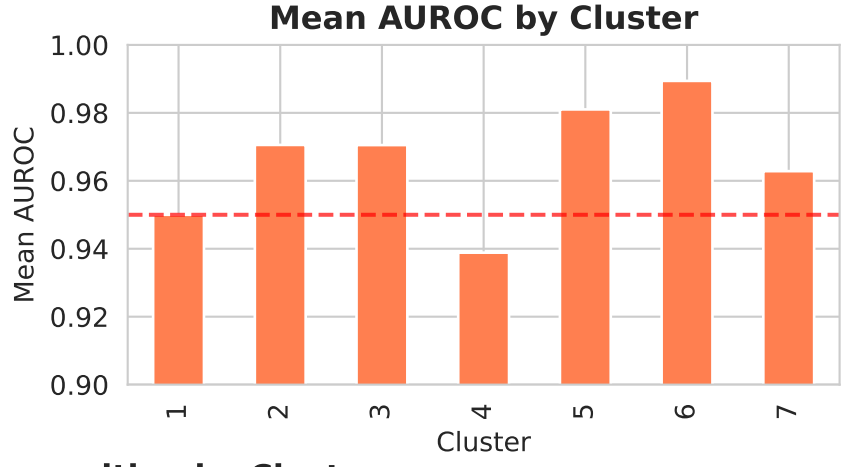
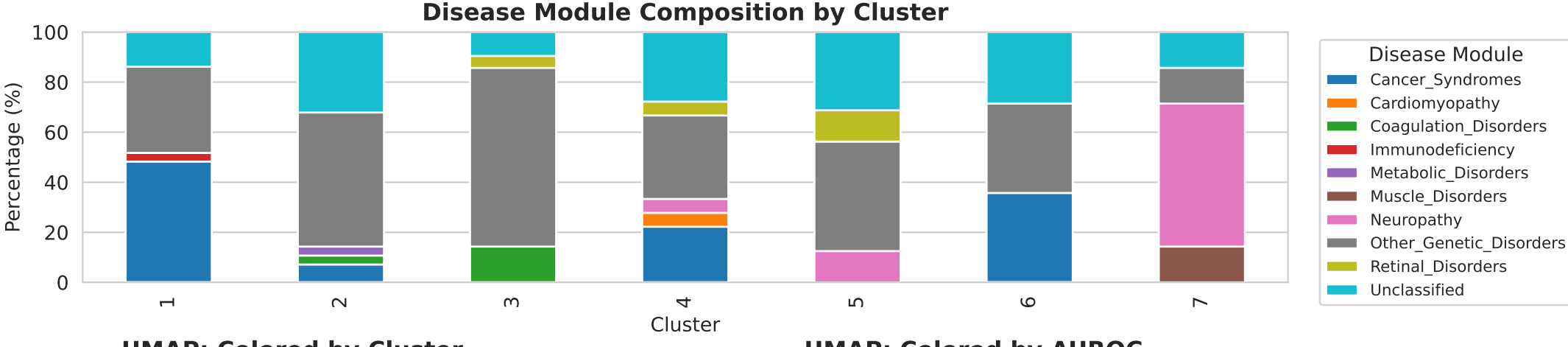
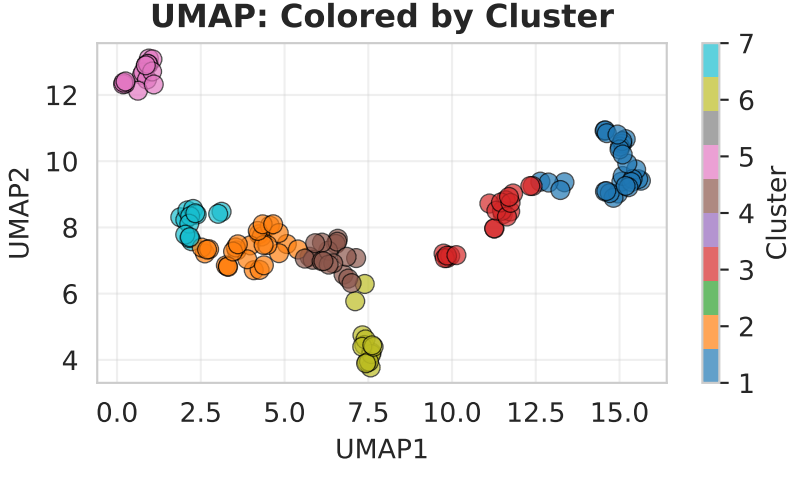
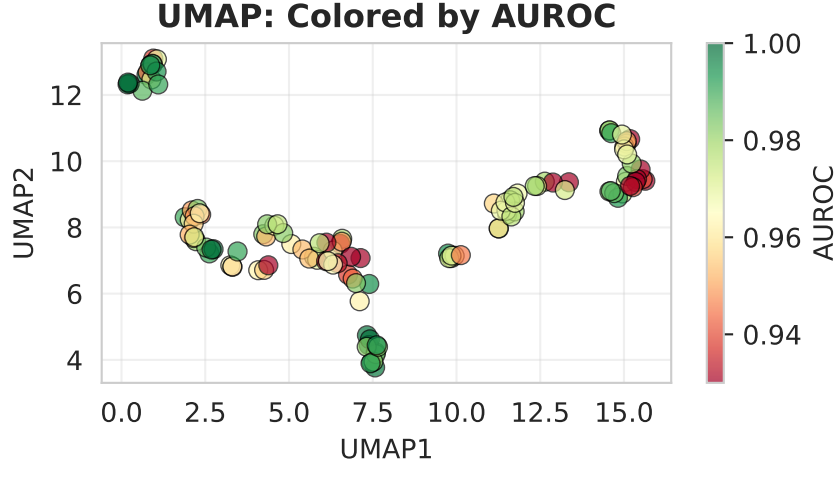
# Comprehensive Disease Biology Analysis - Overview



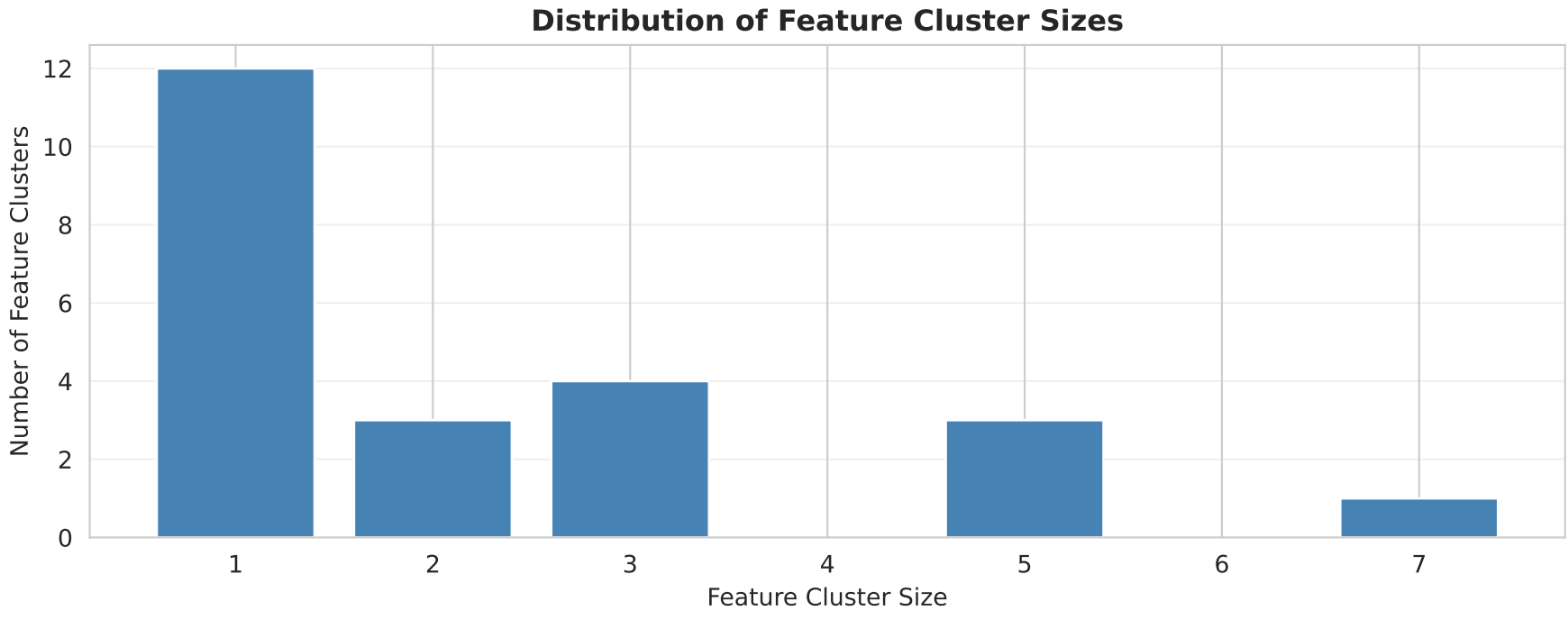



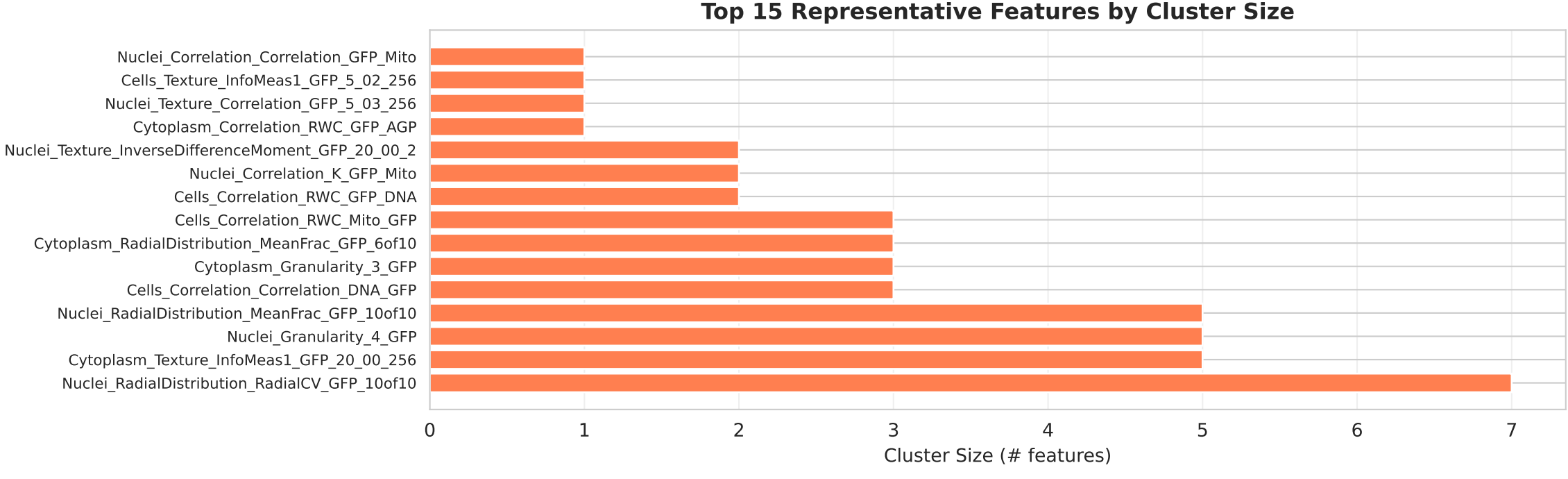



# Gene and Disease Module Patterns



# Feature Correlation Clustering Results



# Cluster 1: Detailed Feature Analysis

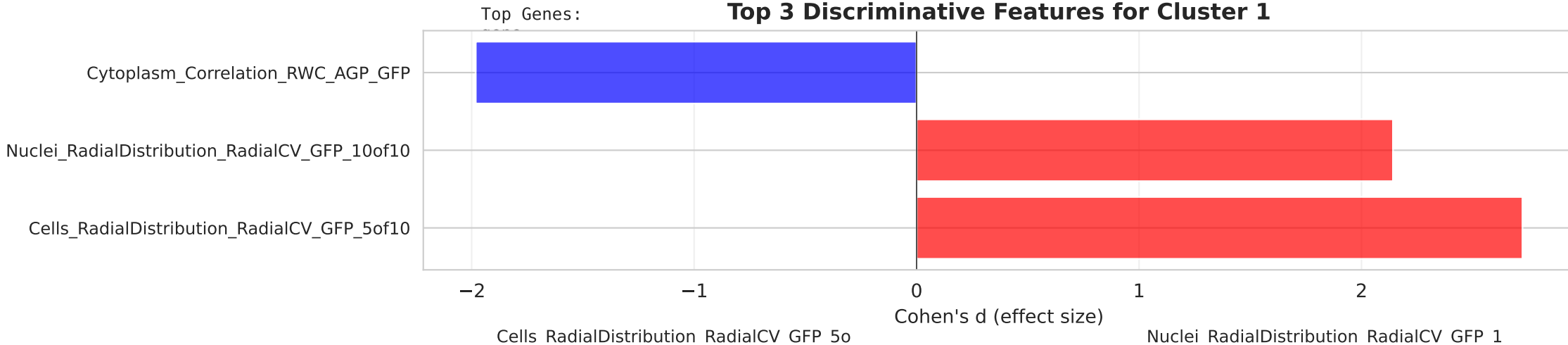
```
CLUSTER 1 ANALYSIS
```

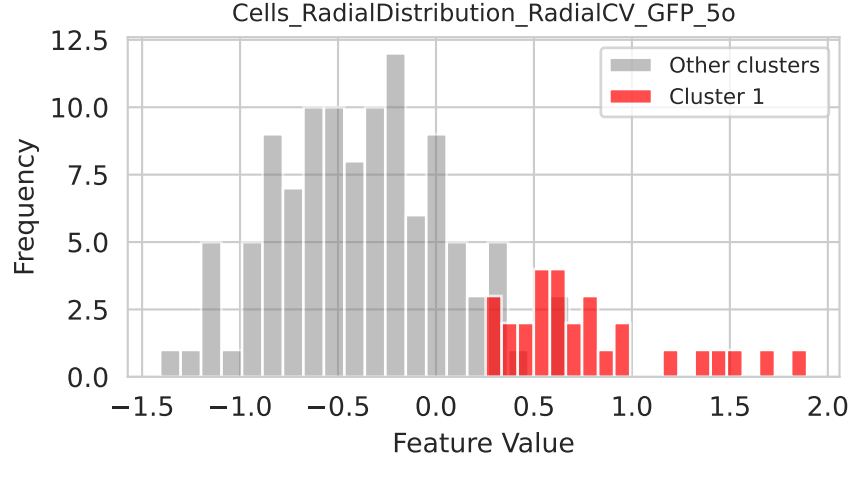
Size: 29 variants
Mean AUROC: 0.950
Altered variants: 22 (75.9%)

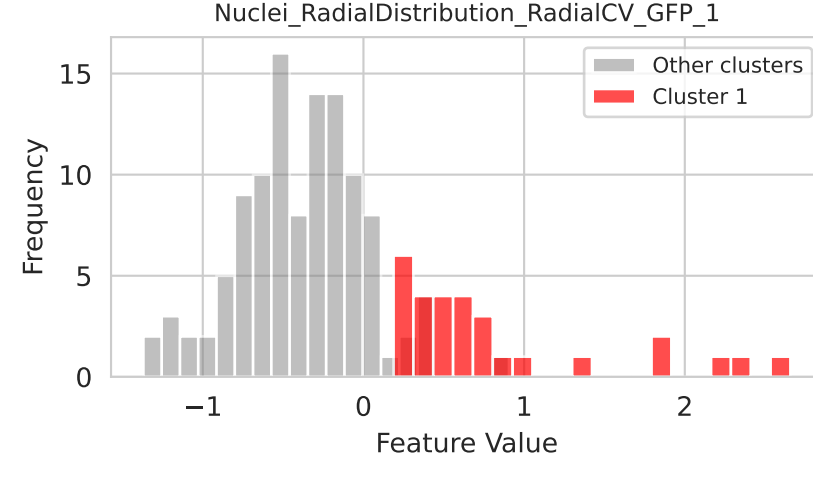
Disease Module Distribution:

| primary_disease_module | |
|---|---|
| Cancer_Syndromes | 14 |
| Other_Genetic_Disorders | 10 |
| Unclassified | 4 |
| Immunodeficiency | 1 |

Top Genes:



# Cluster 2: Detailed Feature Analysis

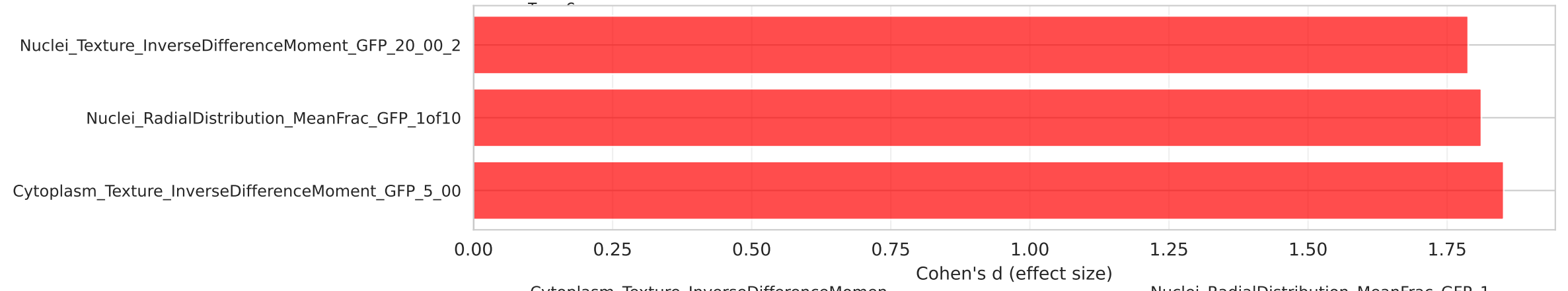
```
CLUSTER 2 ANALYSIS
```

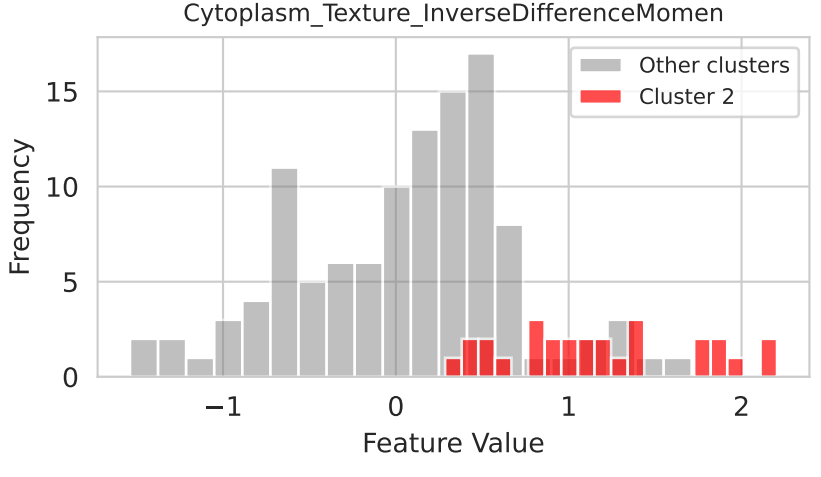
Size: 28 variants
Mean AUROC: 0.971
Altered variants: 16 (57.1%)

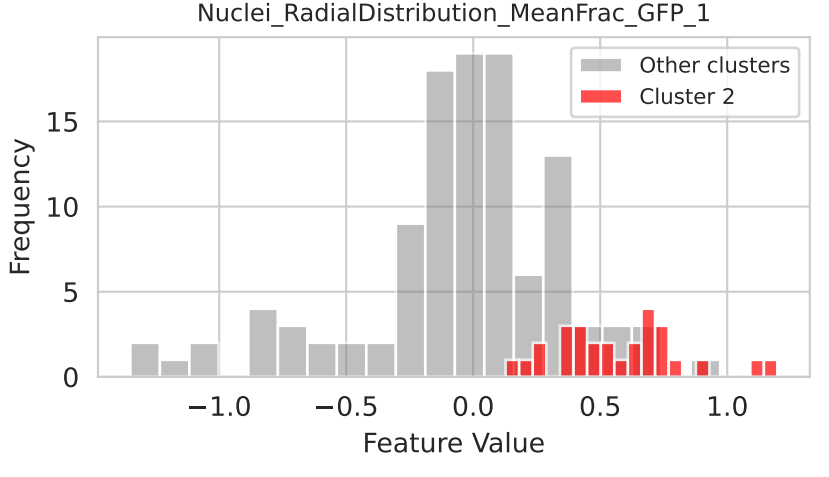
Disease Module Distribution:

| primary_disease_module | |
|---|---|
| Other_Genetic_Disorders | 15 |
| Unclassified | 9 |
| Cancer_Syndromes | 2 |
| Coagulation_Disorders | 1 |
| Metabolic_Disorders | 1 |

## Top 3 Discriminative Features for Cluster 2



# Cluster 3: Detailed Feature Analysis

```
CLUSTER 3 ANALYSIS
```

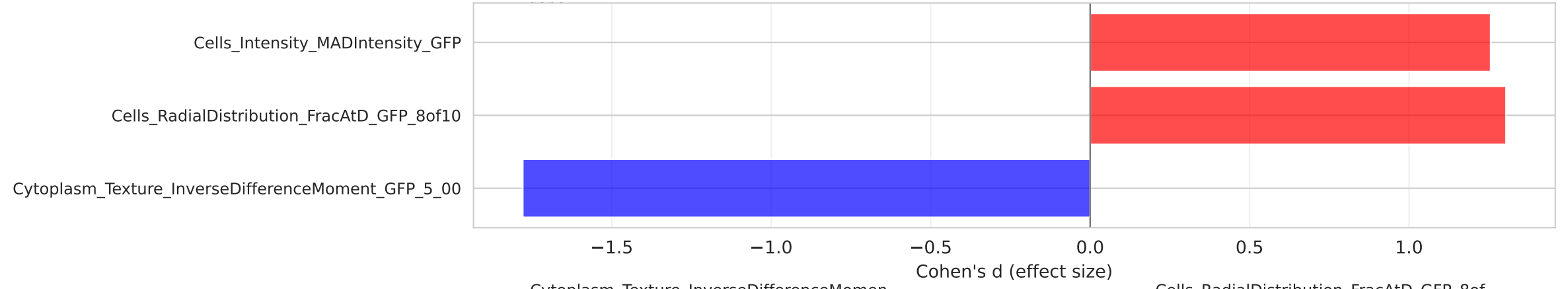
Size: 21 variants
Mean AUROC: 0.971
Altered variants: 16 (76.2%)

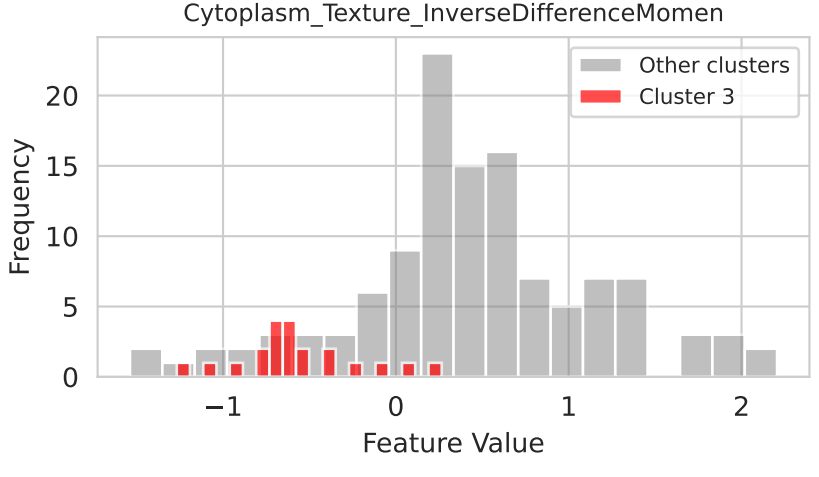
Disease Module Distribution:

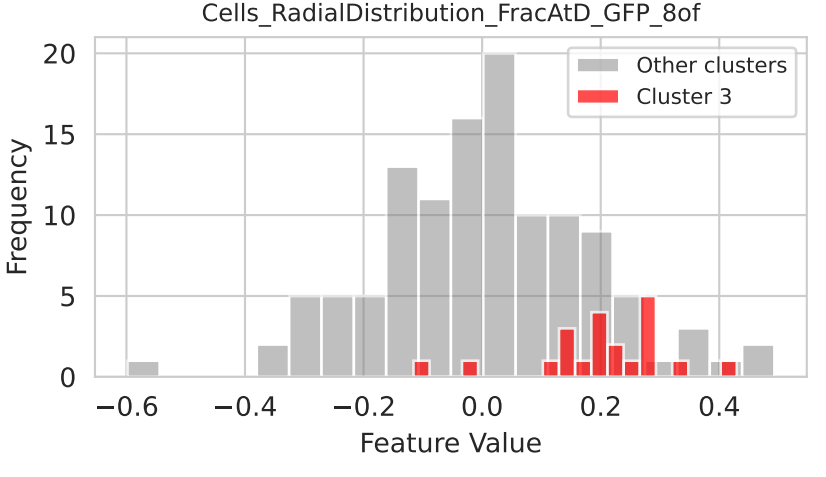
| primary_disease_module | |
|---|---|
| Other_Genetic_Disorders | 15 |
| Coagulation_Disorders | 3 |
| Unclassified | 2 |
| Retinal_Disorders | 1 |

Top Genes:

## Top 3 Discriminative Features for Cluster 3



# Cluster 4: Detailed Feature Analysis

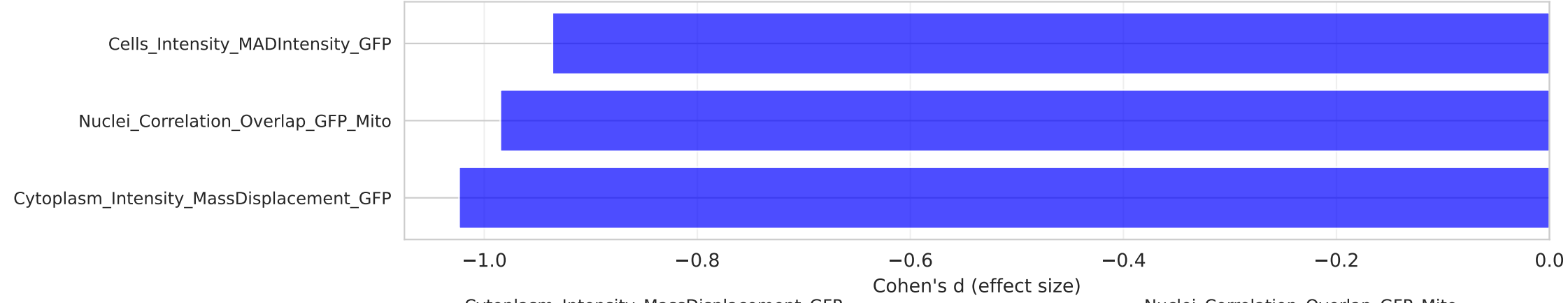
```
CLUSTER 4 ANALYSIS
```

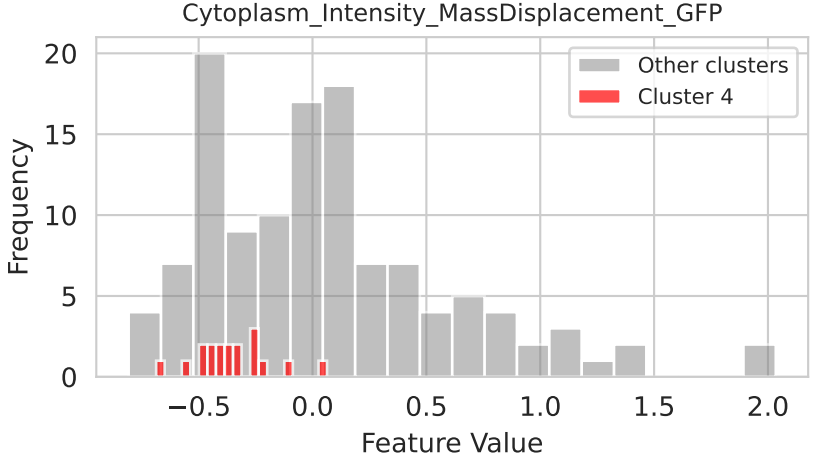
Size: 18 variants
Mean AUROC: 0.939
Altered variants: 11 (61.1%)

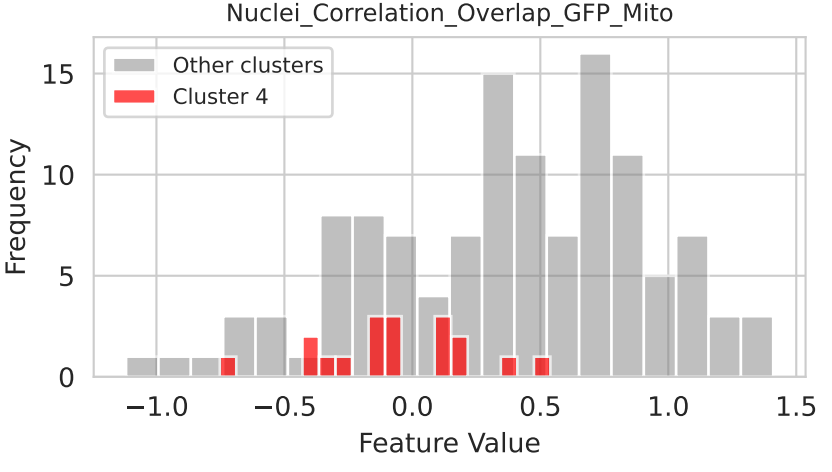
Disease Module Distribution:

| primary_disease_module | |
|---|---|
| Other_Genetic_Disorders | 6 |
| Unclassified | 5 |
| Cancer_Syndromes | 4 |
| Cardiomyopathy | 1 |
| Neuropathy | 1 |
| Retinal_Disorders | |

## Top 3 Discriminative Features for Cluster 4



# Cluster 5: Detailed Feature Analysis

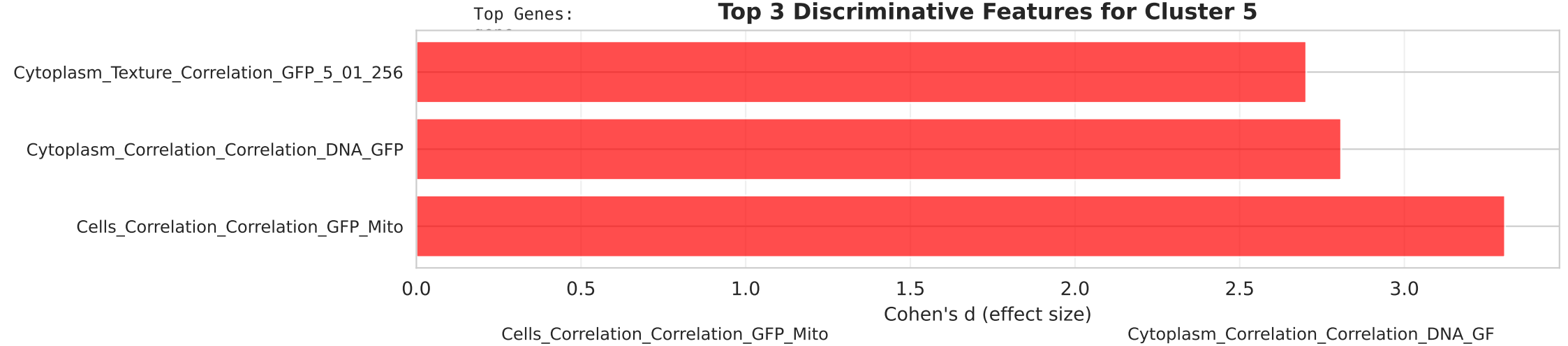
```
CLUSTER 5 ANALYSIS
```

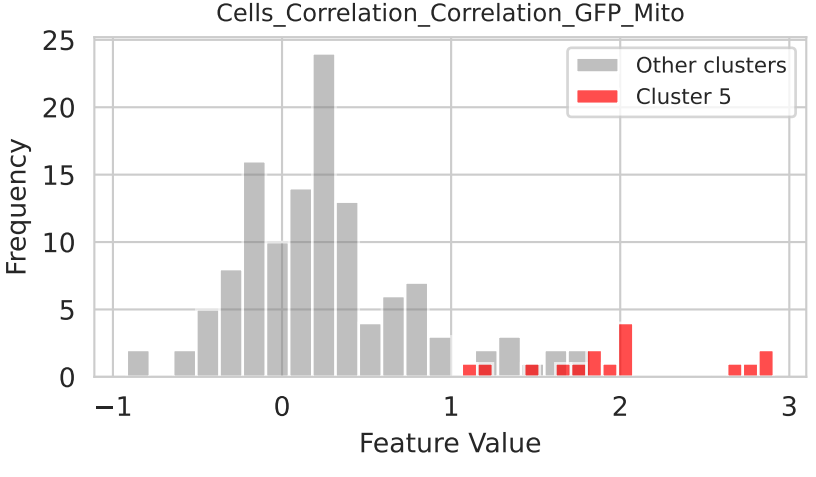
Size: 16 variants
Mean AUROC: 0.981
Altered variants: 14 (87.5%)

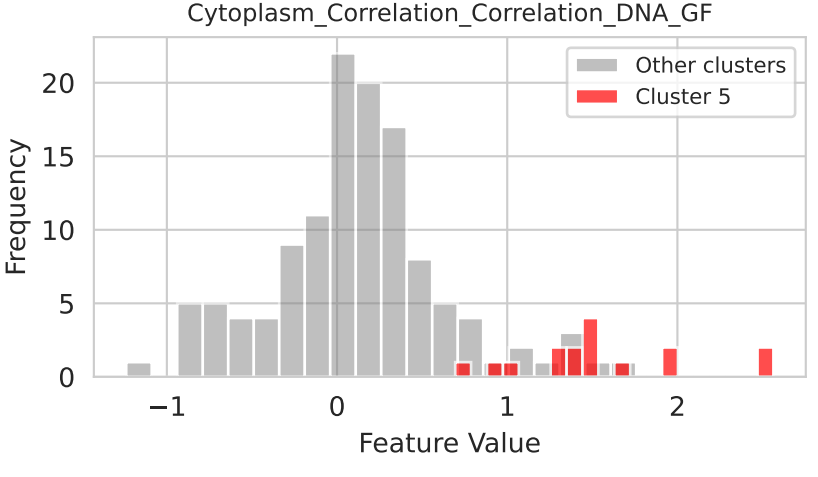
Disease Module Distribution:

| primary_disease_module | |
|---|---|
| Other_Genetic_Disorders | 7 |
| Unclassified | 5 |
| Neuropathy | 2 |
| Retinal_Disorders | 2 |

Top Genes:



# Cluster 6: Detailed Feature Analysis

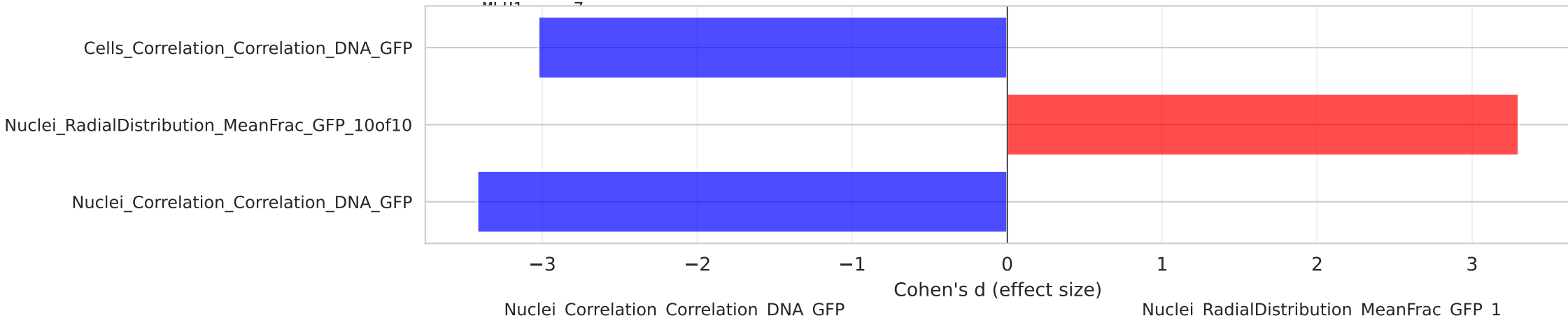
```
CLUSTER 6 ANALYSIS
```

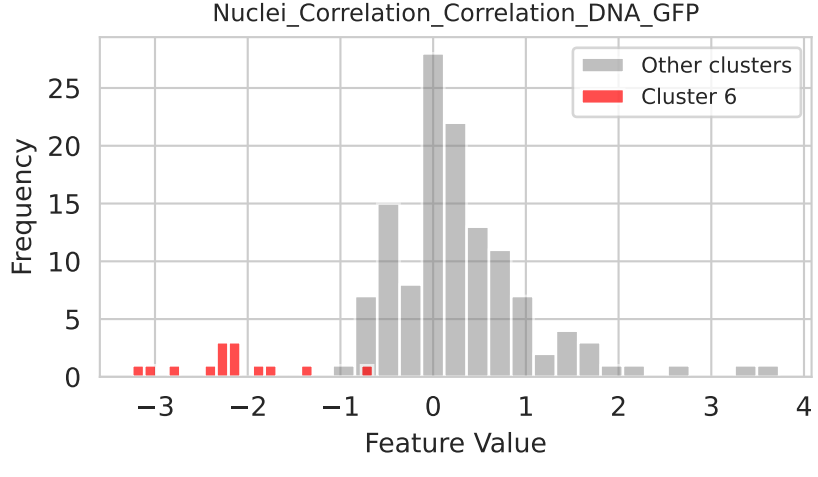
```
Size: 14 variants
Mean AUROC: 0.989
Altered variants: 14 (100.0%)

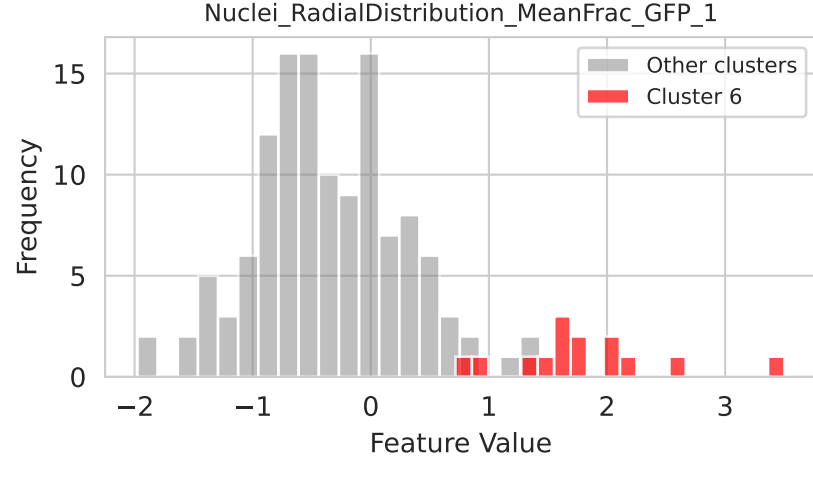
Disease Module Distribution:
primary_disease_module
Other_Genetic_Disorders    5
Cancer_Syndromes           5
Unclassified               4

Top Genes:
gene
```

## Top 3 Discriminative Features for Cluster 6



# Cluster 7: Detailed Feature Analysis

```
CLUSTER 7 ANALYSIS
```

Size: 14 variants
Mean AUROC: 0.963
Altered variants: 8 (57.1%)

Disease Module Distribution:

| primary_disease_module | |
|---|---|
| Neuropathy | 8 |
| Muscle_Disorders | 2 |
| Other_Genetic_Disorders | 2 |
| Unclassified | 2 |

Top Genes:

